

Complete Hadoop 3.3.6 Setup Guide

Windows & Ubuntu

Quick Reference Links

What	Link
Java 8 (Adoptium)	https://adoptium.net/temurin/releases/?version=8
Hadoop 3.3.6	https://hadoop.apache.org/release/3.3.6.html
7-Zip (Windows)	https://www.7-zip.org
winutils (Windows)	https://github.com/cdarlint/winutils
Eclipse IDE	https://www.eclipse.org/downloads/

Java 8 → Hadoop → Configure XML → Format (ONCE) → Start → Use!

WINDOWS SETUP

PHASE 1: Install Java 8

Step 1 — Check if Java is already installed

Open CMD and type:

```
java -version
```

- If you see openjdk version "1.8.0_xxx" → Java 8 already installed! Skip to Step 4.
- If you see Java 11, 17, or 21 → You still need Java 8 specifically for Hadoop. Install it alongside.
- If you see 'java is not recognized' → Java not installed. Continue with Step 2.

Step 2 — Download Java 8

Go to: <https://adoptium.net/temurin/releases/?version=8>

Select: Version=8 | OS=Windows | Architecture=x64 | Package=JDK

Click Download → downloads a .msi file

Step 2 — Install Java

Run the .msi file → keep clicking Next → Next → Install

Default install path:

```
C:\Program Files\Eclipse Adoptium\jdk-8.0.xxx-hotspot
```

Step 3 — Set Environment Variables

Press Win + S → type 'Environment Variables' → Click 'Edit the system environment variables' → Click 'Environment Variables' button

Create New System Variable:

```
Variable Name → JAVA_HOME Variable Value → C:\Program Files\Eclipse  
Adoptium\jdk-8.0.xxx-hotspot
```

Edit PATH → Click New → Add:

```
%JAVA_HOME%\bin
```

Click OK → OK → OK

- Replace xxx with your actual version number shown in the folder

Step 4 — Verify Java

```
java -version
```

- Should show: openjdk version "1.8.0_xxx"

PHASE 2: Download & Install Hadoop 3.3.6

Step 1 — Download Hadoop

Go to: <https://hadoop.apache.org/release/3.3.6.html>

Click Download tar.gz (first green button)

Step 2 — Install 7-Zip

Go to: <https://www.7-zip.org> → Download and install it

Step 3 — Extract Hadoop

Right click .tar.gz → 7-Zip → Extract Here (gives .tar file)

Right click .tar → 7-Zip → Extract Here again

Rename folder to hadoop → Move to C:\ → Final path: C:\hadoop

Step 4 — Download winutils (Windows fix)

Go to: <https://github.com/cdarlint/winutils>

Open folder hadoop-3.3.6 → bin → Download winutils.exe and hadoop.dll

```
Paste both files into: C:\hadoop\bin\
```

Step 5 — Set Hadoop Environment Variables

Create New System Variables:

```
Variable Name → HADOOP_HOME Variable Value → C:\hadoop Variable Name → HADOOP_BIN_PATH  
Variable Value → C:\hadoop\bin
```

Edit PATH → Add both:

```
%HADOOP_HOME%\bin %HADOOP_HOME%\sbin
```

Click OK → OK → OK

Step 6 — Verify Hadoop

```
hadoop version
```

■ Should show: Hadoop 3.3.6

PHASE 3: Configure Hadoop (XML Files)

All files are in C:\hadoop\etc\hadoop\ — Open each with Notepad

■ File 1: core-site.xml

```
<configuration> <property> <name>fs.defaultFS</name> <value>hdfs://localhost:9000</value>  
</property> </configuration>
```

■ File 2: hdfs-site.xml

First manually create these 2 folders:

```
C:\hadoop\data\namenode C:\hadoop\data\datanode
```

```
<configuration> <property> <name>dfs.replication</name> <value>1</value> </property>  
<property> <name>dfs.namenode.name.dir</name> <value>C:\hadoop\data\namenode</value>  
</property> <property> <name>dfs.datanode.data.dir</name>  
<value>C:\hadoop\data\datanode</value> </property> </configuration>
```

■ File 3: mapred-site.xml

```
<configuration> <property> <name>mapreduce.framework.name</name> <value>yarn</value>  
</property> </configuration>
```

■ File 4: yarn-site.xml

```
<configuration> <property> <name>yarn.nodemanager.aux-services</name>  
<value>mapreduce_shuffle</value> </property> </configuration>
```

■ File 5: hadoop-env.cmd

Open C:\hadoop\etc\hadoop\hadoop-env.cmd

Find: set JAVA_HOME=%JAVA_HOME% → Replace with:

```
set JAVA_HOME=C:\PROGRA~1\Eclipse Adoptium\jdk-8.0.xxx-hotspot
```

■ Use PROGRA~1 instead of Program Files to avoid space errors

PHASE 4: First Time Setup (ONCE ONLY!)

Open CMD as Administrator and run:

```
hdfs namenode -format
```

■ Run this ONLY ONCE ever! Running again = all HDFS data deleted!

PHASE 5: Start Hadoop (Every Time)

Always open CMD as Administrator!

Start HDFS:

```
start-dfs.cmd
```

→ 2 black windows open (Namenode + Datanode) ■

Start YARN:

```
start-yarn.cmd
```

→ 2 more windows open (ResourceManager + NodeManager) ■

Total 4 black windows should be open!

Verify in Browser:

Service	URL
HDFS UI	http://localhost:9870
YARN UI	http://localhost:8088

PHASE 6: Run WordCount Example

Step 1 — Create C:\file1.txt with content:

```
Hello Hadoop Hello World Hadoop is great
```

Step 2 — Create HDFS input folder

```
hdfs dfs -mkdir /input
```

Step 3 — Upload file to HDFS

```
hdfs dfs -copyFromLocal C:\file1.txt /input
```

Step 4 — Verify upload

```
hdfs dfs -ls /input
```

Step 5 — Run WordCount job

```
yarn jar C:\hadoop\share\hadoop\mapreduce\hadoop-mapreduce-examples-3.3.6.jar wordcount /input /output
```

Step 6 — See output

```
hdfs dfs -cat /output/part-r-00000
```

Result: Hadoop=2, Hello=2, World=1, great=1, is=1

PHASE 7: Stop Hadoop (When Done)

```
stop-yarn.cmd
```

```
stop-dfs.cmd
```

Windows Troubleshooting

Problem	Fix
JAVA_HOME not set	Use PROGRA~1 in hadoop-env.cmd
Namenode won't start	Delete C:\hadoop\data, recreate folders, run format again
Output dir exists	Run: hdfs dfs -rm -r /output first
winutils error	Check winutils.exe is in C:\hadoop\bin
Port conflict	Restart PC and try again
Windows not opening	Open CMD as Administrator

UBUNTU SETUP

PHASE 1: Install Java 8

Step 1 — Check if Java is already installed

Open Terminal and type:

```
java -version
```

- If you see openjdk version "1.8.0_xxx" → Java 8 already installed! Skip to Phase 2.
- If you see Java 11, 17, or 21 → You still need Java 8 for Hadoop. Install it alongside.
- If you see 'command not found' → Java not installed. Continue with Step 2.

Step 2 — Install Java 8

Open Terminal and run:

```
sudo apt update
```

```
sudo apt install openjdk-8-jdk -y
```

Verify:

```
java -version
```

- Should show: openjdk version "1.8.0_xxx"

PHASE 2: Set Environment Variables

Open the bashrc file:

```
nano ~/.bashrc
```

Add these lines at the bottom of the file:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64 export HADOOP_HOME=~/.hadoop export  
PATH=$HADOOP_HOME/bin:$HADOOP_HOME/sbin:$PATH
```

Save file: Ctrl+X → Y → Enter

Apply changes:

```
source ~/.bashrc
```

PHASE 3: Download & Extract Hadoop 3.3.6

Download Hadoop directly:

```
wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
```

Extract it:

```
tar -xvzf hadoop-3.3.6.tar.gz
```

Rename and move:

```
mv hadoop-3.3.6 hadoop
```

PHASE 4: Configure Hadoop (XML Files)

All files are in ~/.hadoop/etc/hadoop/ — Edit using nano

■ File 1: hadoop-env.sh

```
nano ~/.hadoop/etc/hadoop/hadoop-env.sh
```

Add this line:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
```

■ File 2: core-site.xml

```
nano ~/hadoop/etc/hadoop/core-site.xml
```

```
<configuration> <property> <name>fs.defaultFS</name> <value>hdfs://localhost:9000</value>
</property> </configuration>
```

■ File 3: hdfs-site.xml

```
nano ~/hadoop/etc/hadoop/hdfs-site.xml
```

```
<configuration> <property> <name>dfs.replication</name> <value>1</value> </property>
<property> <name>dfs.namenode.name.dir</name>
<value>file:///home/vboxuser/hadoopdata/namenode</value> </property> <property>
<name>dfs.datanode.data.dir</name>
<value>file:///home/vboxuser/hadoopdata/datanode</value> </property> </configuration>
```

■ Replace 'vboxuser' with your actual Ubuntu username

■ File 4: mapred-site.xml

```
nano ~/hadoop/etc/hadoop/mapred-site.xml
```

```
<configuration> <property> <name>mapreduce.framework.name</name> <value>yarn</value>
</property> </configuration>
```

■ File 5: yarn-site.xml

```
nano ~/hadoop/etc/hadoop/yarn-site.xml
```

```
<configuration> <property> <name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value> </property> </configuration>
```

PHASE 5: Setup SSH (Required for Hadoop)

Generate SSH key:

```
ssh-keygen -t rsa -P ""
```

Add to authorized keys:

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

Set permissions:

```
chmod 0600 ~/.ssh/authorized_keys
```

Test SSH:

```
ssh localhost
```

■ Should connect without asking for password

PHASE 6: Create Hadoop Data Directories

```
mkdir -p ~/hadoopdata/namenode
```

```
mkdir -p ~/hadoopdata/datanode
```

PHASE 7: Format NameNode (ONCE ONLY!)

```
hdfs namenode -format
```

■ Run this ONLY ONCE ever! Running again = all HDFS data deleted!

PHASE 8: Start Hadoop (Every Time)

Start HDFS:

```
start-dfs.sh
```

Start YARN:

```
start-yarn.sh
```

Verify all services running:

```
jps
```

■ Should show: NameNode, DataNode, ResourceManager, NodeManager, Jps

Verify in Browser:

Service	URL
HDFS UI	http://localhost:9870
YARN UI	http://localhost:8088

PHASE 9: Run WordCount Example

Step 1 — Create input file:

```
echo "hello hadoop hello world" > file1.txt
```

Step 2 — Create HDFS directory and upload:

```
hdfs dfs -mkdir /input
```

```
hdfs dfs -put file1.txt /input
```

Step 3 — Run WordCount:

```
hadoop jar ~/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.6.jar wordcount /input /output
```

Step 4 — View output:

```
hdfs dfs -cat /output/part-r-00000
```

PHASE 10: Stop Hadoop (When Done)

```
stop-yarn.sh
```

```
stop-dfs.sh
```

Ubuntu Troubleshooting

Problem	Fix
SSH connection refused	Run: <code>sudo service ssh start</code>
JAVA_HOME not set	Check <code>~/.bashrc</code> and run: <code>source ~/.bashrc</code>
Namenode won't start	Delete <code>~/hadoopdata</code> , recreate it, run format again
Output dir exists	Run: <code>hdfs dfs -rm -r /output</code> first
Permission denied	Run: <code>chmod 0600 ~/.ssh/authorized_keys</code>
Command not found	Run: <code>source ~/.bashrc</code> to reload env variables